Title : Question Paper

FF No. 868

Reg.No.

Bansilal Ramnath Agarwal Charitable Trust's
VISHWAKARMA INSTITUTE OF TECHNOLOGY, PUNE – 411037.
(An Autonomous Institute Affiliated to Savitribai Phule Pune University)

Examination: ESE

Year: TY **Branch:** Computer

Subject: Design and Analysis of Algorithms **Subject Code:** CS 3205

Max. Marks:60 **Total Pages of Question Paper: 3**

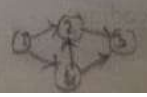
Day & Date 23/11/2023. **Time:** 10:30am to 12:30pm

Instructions to Candidate

1. Figures to the right indicate full marks.
2. Try to avoid using informal arguments. They won't get any credits. Try to write formal, concrete, and concise answers.

Q. N.	CO No	BT* No		Max marks
Q. 1.				
Q.1. (a)	3	2, 3	For the following two recurrences give best possible asymptotic upper bound on T(n) as a function of n. Express your answer using Big Oh asymptotic notation. (1 mark for correct final answer and 2 marks for the correct approach) (1) T(n) = T (n-1) +n^2. (2) T(n) = 9 T (n/3) + n^2.	6 marks
Q.1. (b)			Attempt Any ONE of the following.	
Q.1. (b)(i)	3	2, 3	Let F_n denotes n th Fibonacci number. (that is F_1=1, F_2=1 and for n>2 F_n = F_{n-1}+F_{n-2}). Prove that there exists an absolute constant a >1 such that F_n is Omega (a^n). What is a best possible value of the constant a one can choose?	4 marks
Q.1. (b)(ii)	3	1,3	Formally define the Big Oh asymptotic notation. When do we say a relation is transitive? Prove that the Big Oh asymptotic notation induces a transitive relation on set of functions from the set of the natural numbers to the set of the natural numbers. (Note that informal definitions and arguments do not carry any credits)	
Q. 2.				
Q. 2. (a)	1, 2, 3	1, 3	A. Let f(n) denotes number of binary trees on n nodes. Give a recurrence relation for f(n). (2 marks) B. A full binary tree is a binary tree in which every node has either zero or two children. Let B_n denote the number of full binary trees on n nodes. (1) Give a recurrence relation for B_n (2) Prove that if n is an even number, then B_n =0 (3) Using the recurrence above or otherwise determine B_n for all n<=7.	8 marks
Q. 2. (b)	4, 6	1	Give prover-verifier based definition of complexity class NP and give two concrete examples of problems characterizing NP class.	2 marks
Q.3.				
Q. 3. (a)	2, 3	2	Consider a text over alphabet {a,b,c,d,e,f,g} relative frequencies of the characters are 5, 7, 8, 16, 18, 22, 24 respectively. Construct Huffman tree. Find Huffman encoding for each of the seven characters. Suppose original text contains total 1,00,000 characters what will be size of the text in bits after encoding. (5 marks for Huffman tree and correct codes, 1 mark for correct text size after encoding)	6 marks

Q. 3. (b)			Attempt any **ONE** of the following	**6 marks**
Q. 3. (b) (i)	1,2,3,6	3, 4	Given integers n and k, and numbers p_1, … , p_n ∈ [0, 1] (that is 0 <= p_i <= 1 for every i). You want to determine the probability of obtaining exactly k heads when n biased coins are tossed independently at random, where p_i is the probability that the ith coin comes up heads. Give an O(n^2) time algorithm for this task with proper Pseudo code. Give an argument justifying correctness of the algorithm and the time complexity. Assume you can multiply and add two numbers in [0, 1] in O(1) time.	
Q. 3. (b) (ii)	1, 2, 3, 6	3, 4	Let S be a set of n positive integers each of which is less than k. You are given set S and a natural number b, such that b < nk and goal is to check if there exist a subset A of S such that the addition of all the elements in the subset A is b. Give an algorithm which has time complexity bounded above by O((nk)^c) for an absolute constant c. Give detailed pseudo-code and the complexity analysis. (Hint: Note similarity with coin change problem)	
Q.4.	1, 2,3 5,6	4, 6	A vertex cover of a graph G(V,E) is a set of vertices S such that every edge of G has at least one end point in S. A tree is an undirected connected graph which doesn't have any cycles. Given a tree T on n vertices as input, give an efficient algorithm for finding size of a smallest sized vertex cover for T. Give formal pseudocode and analyze the time complexity of the algorithm. Full credit will be given only for the correct algorithm whose time complexity is asymptotically equal to the time complexity of the best algorithm for the problem.	**8 marks**
Q.5.	1, 2, 3, 6	4, 6	Suppose n circles are placed in 2-D plane. You are given coordinates of center and radius for each of the circles. A ray is a line segment starting at origin (0,0) and extending all the way up-to infinity in any given direction. Give an efficient algorithm to find the minimum number of rays needed to cover all the circles. We say a ray covers a circle if the ray passes through the circle non-trivially or the ray is tangential to the circle. Prove the correctness of your algorithm. Assume that you can do any geometric task such as finding intersection point(s) of a line and a circle, finding a point of contact of a circle and a tangent drawn from point outside the circle etc in O(1) time. (3 marks for the correct algorithm, 5 marks for the correctness proof of the algorithm) For full credit time complexity of your algorithm must be asymptotically equal to the time complexity of the best algorithm for the problem.	**8 marks**
Q.6.				
Q. 6. (a)	1, 2, 3, 6	4, 6	You are given two **sorted** arrays A and B each containing n distinct integers and a natural number r <= 2n as input. Assume that there is no common element in A and B. Let C be set of all elements either in A or B (that is C is union of A and B). For an element p in array C rank(p) in C is a total number of elements in C which are less than or equal to p. (E.g. if C = {2, 7, 1, 9} then rank(7) in C is 3 as elements 1, 2, 7 in C are less than or equal to 7) . Give an efficient algorithm to find an element p in C with rank(p) = r. Give proper Pseudo-code for the algorithm. Analyze the time complexity of the algorithm. For full credit time complexity of your algorithm must be asymptotically equal to the time complexity of the best algorithm for the problem.	**7 marks**
Q. 6. (b)	1, 2, 3	3, 4	You are given directed graph G(V,E), vertices s, t in V and a natural number k as input. Give an efficient algorithm to compute number of distinct directed paths of length k from vertex s to vertex t in G. Give proper Pseudocode of the algorithm and analyze its time complexity. For full credit time complexity of your algorithm must be asymptotically equal to the time complexity of the best algorithm for the problem.	**5 marks**

CO Statements:

1. To formulate computational problems in abstract and mathematically precise manner